CS264 Computer Networks

Network Layer

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Acknowledgement: Some of the slide contents are taken from the slides shared by the author of the book Computer Networking: A Top-Down Approach 9th Edition, Jim Kurose, Keith Ross. Some of the figures are also taken from the same book. Authors are sincerely acknowledged

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Internet

A massive, global "network of networks" that links billions of computing devices worldwide using shared communication rules

Goal is to achieve data communication

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Networks are complex, with many "pieces"



Billions of connected computing *devices*:
- *hosts* = *end systems*
- running *network apps* at Internet's "edge"

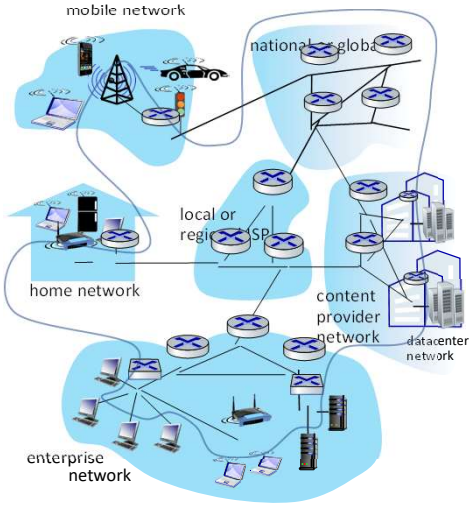


Packet switches: forward packets (chunks of data)
- *routers*, *switches*



Communication links
- fiber, copper, radio, satellite
- transmission rate: *bandwidth*

Networks
- collection of devices, routers, links: managed by an organization



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A Complex System - Internet

- **Internet is pervasive today in day to day life**
- **Need to design, maintain and provide data communication service**
- **Natural way to handle complex systems**
  - **Breakdown the complex process into smaller, independent subtasks**
    - **Provided handling smaller ones is easier**
  - **Brings modularity**
  - **True for the study of the system also**
- **Computer Network uses a "layered approach"**
  - **Each "layer" implements a service**
    - **Use its internal layer action**
    - **Relying on services provided by the layer below**
    - **Provides a service to a layer above**

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Layered Approach

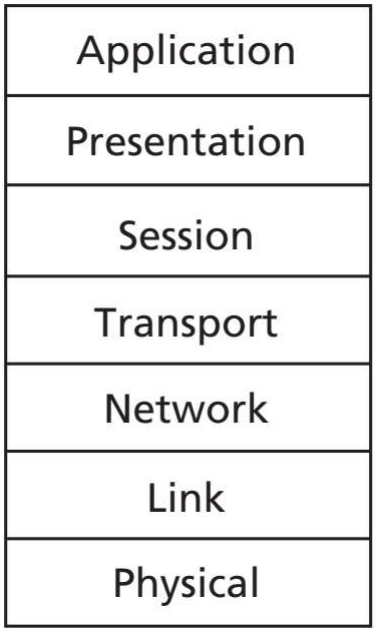
- Computer Network uses a "layered approach"
  - Each "layer" implements a service
    - Use its internal layer action
    - Relying on services provided by the layer below
    - Provides a service to a layer above
- Activities rather responsibilities of a "layer" is defined using protocols
- Layer
  - Protocols
  - Hardware
  - Software that implement the protocol
- Protocol layer can be implemented in software or a combination of software and hardware

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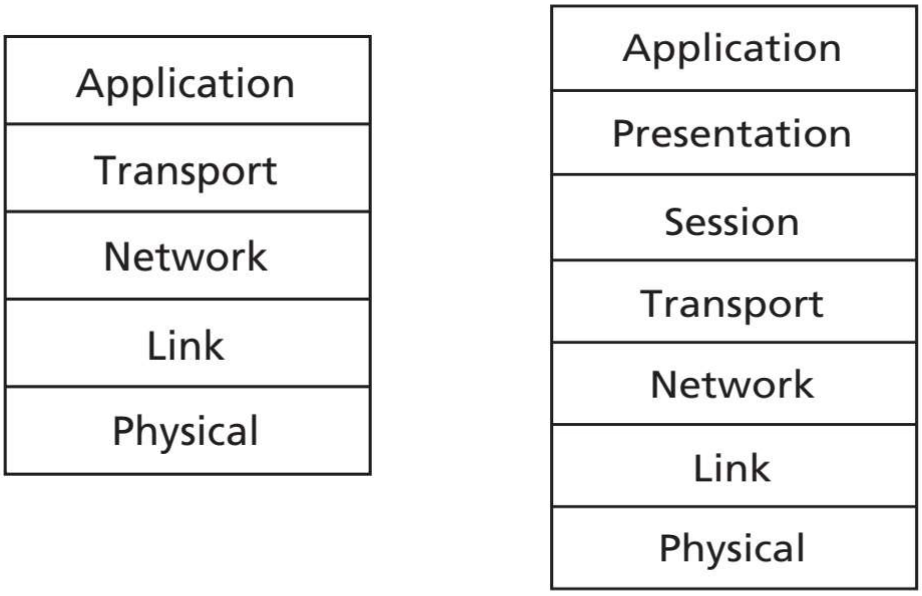
Layered Approach

- Some standardization of protocol stack is necessary to allow devices to communicate
  - Agree or disagree?
- International Organization for Standardization (ISO)
  - Conceptual seven layers framework
  - OSI (Open System Interconnection) reference model



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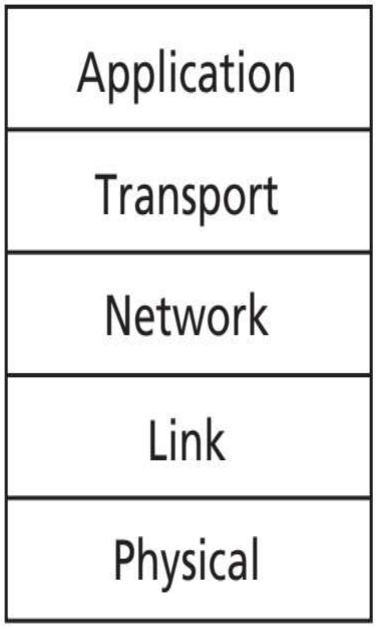
Internet Protocol Stack



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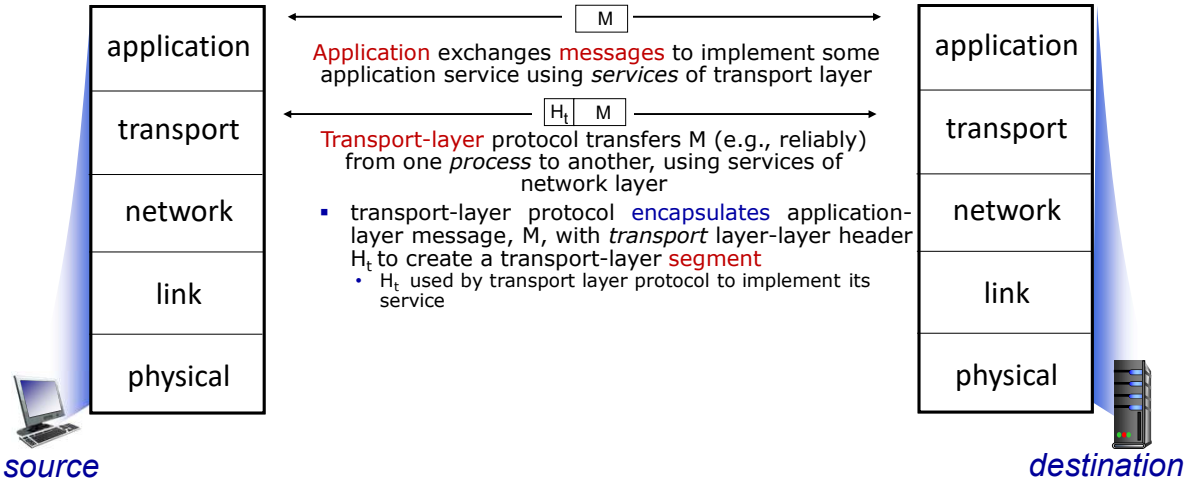
Internet Protocol Stack

- **Application:** Supporting network applications
  - HTTP, IMAP, SMTP, DNS
- **Transport:** Process-process data transfer
  - TCP, UDP
- **Network:** Routing of datagrams from source to destination:
  - IP, routing protocols
- **Link:** data transfer between neighboring network elements
  - Ethernet, 802.11 (WiFi), PPP
- **Physical:** bits "on the wire"



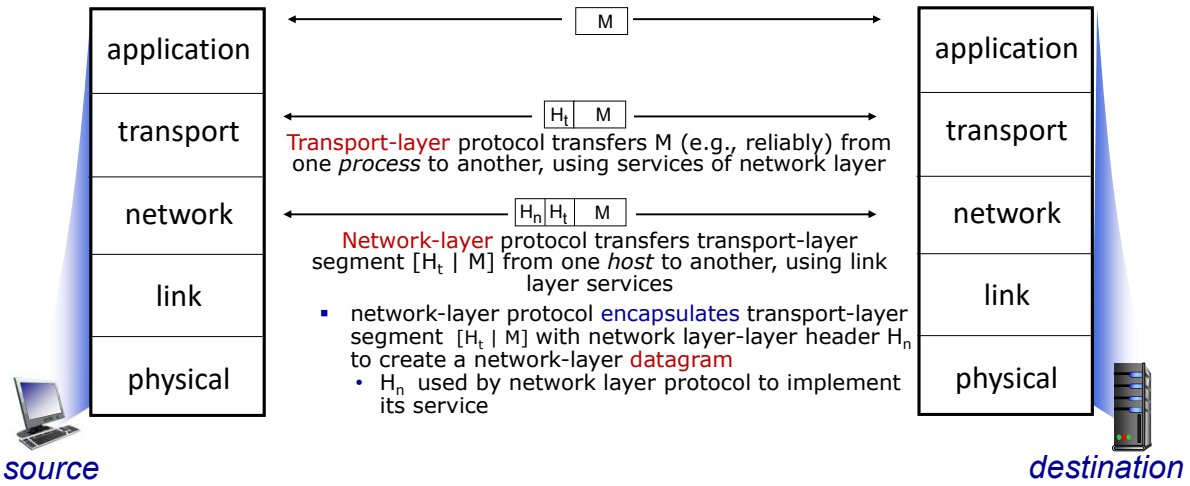
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Services, Layering and Encapsulation



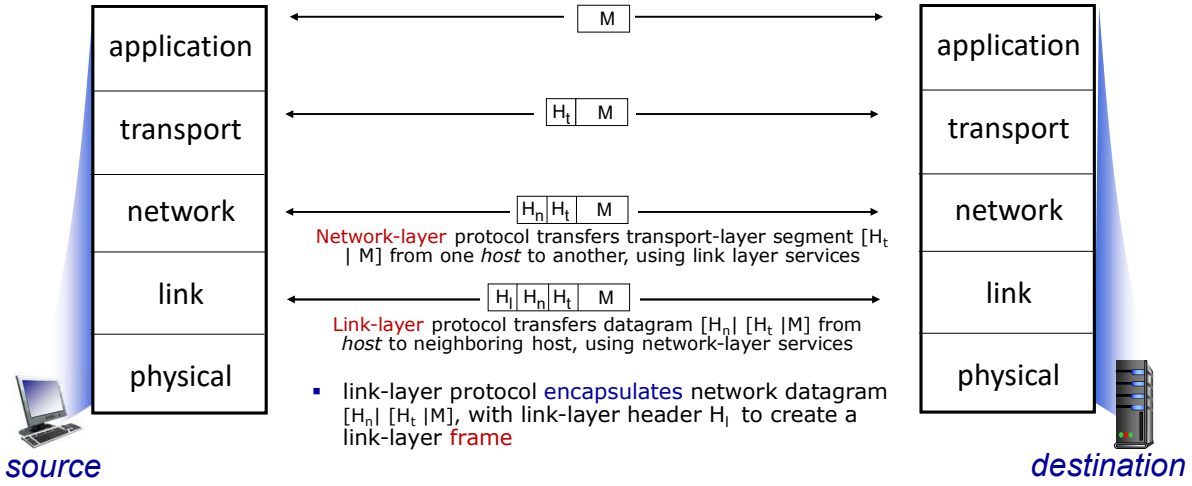
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Services, Layering and Encapsulation



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Services, Layering and Encapsulation

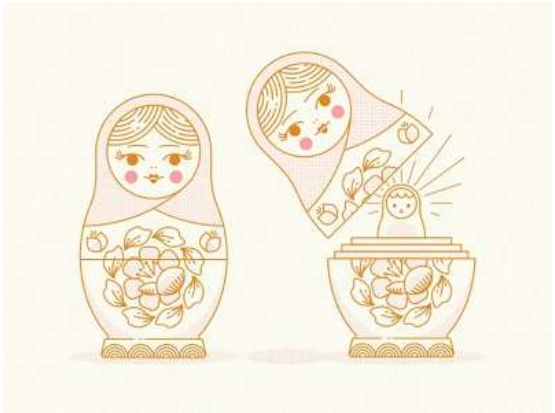


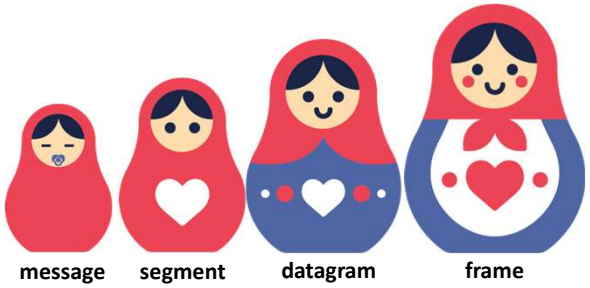
- link-layer protocol encapsulates network datagram [H_n| [H_t |M], with link-layer header H_l to create a link-layer frame

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Encapsulation

Matryoshka dolls (stacking dolls)

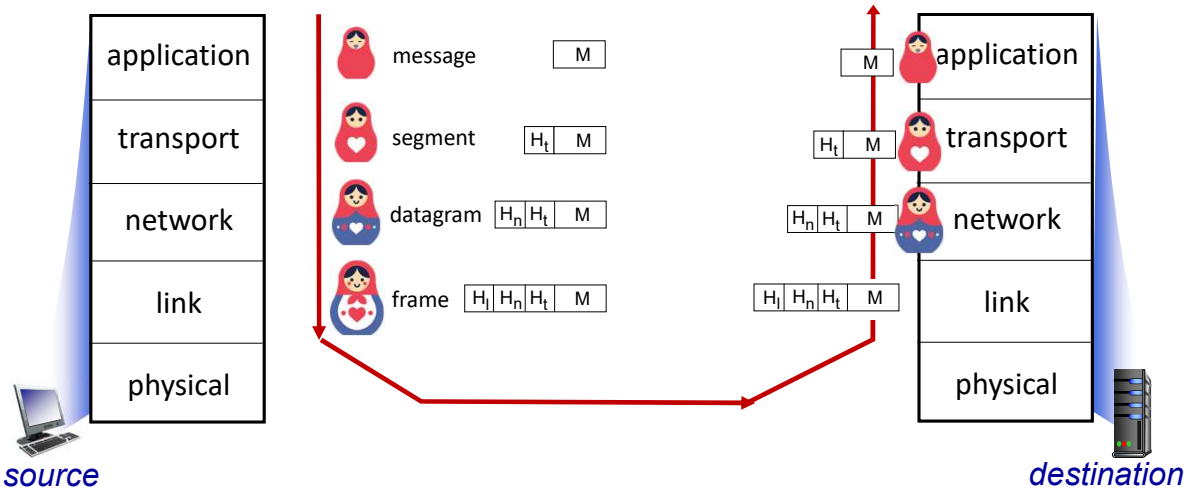




Credit: https://dribbble.com/shots/7182188-Babushka-Boi

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Services, Layering and Encapsulation



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Already Covered

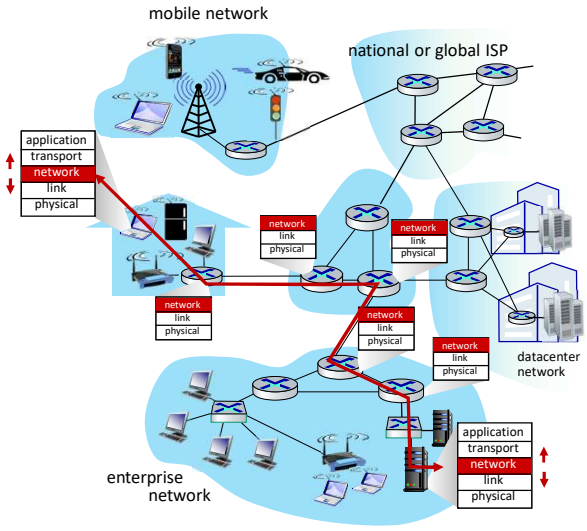
- **Application layer**
- **Transport Layer**

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Network Layer

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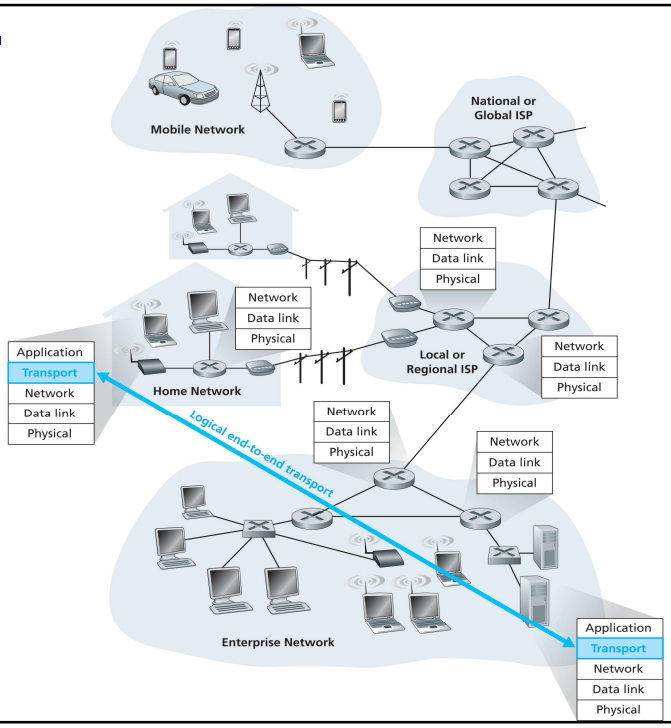
Network Layer



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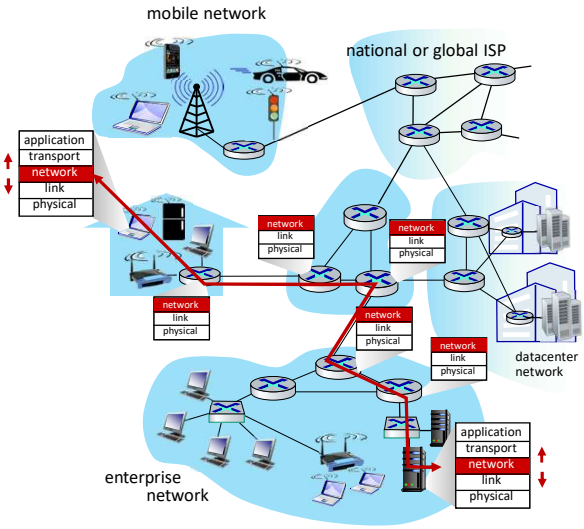
Transport Layer

- Logical communication between the processes



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Network Layer



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Network Layer

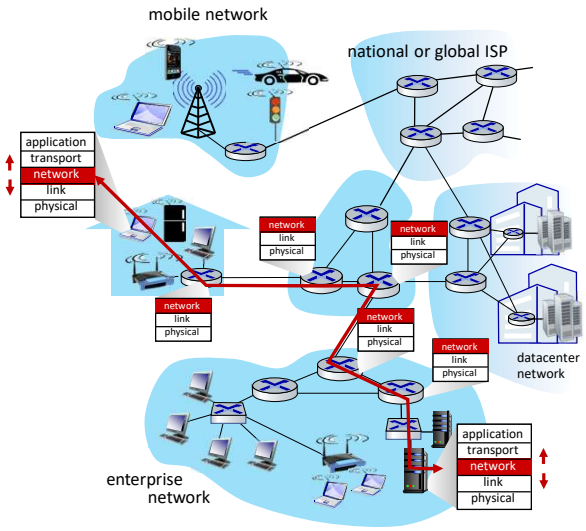
- Transport segment from sending host to receiving host
- Routers
- Network layer protocols in every internet devices
  - Hosts
  - Routers

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Routers-Network Layer

- Examines header fields in all IP datagrams passing through it
- Moves datagrams from input ports to output ports to transfer datagrams along end-end path



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Network Layer Functions

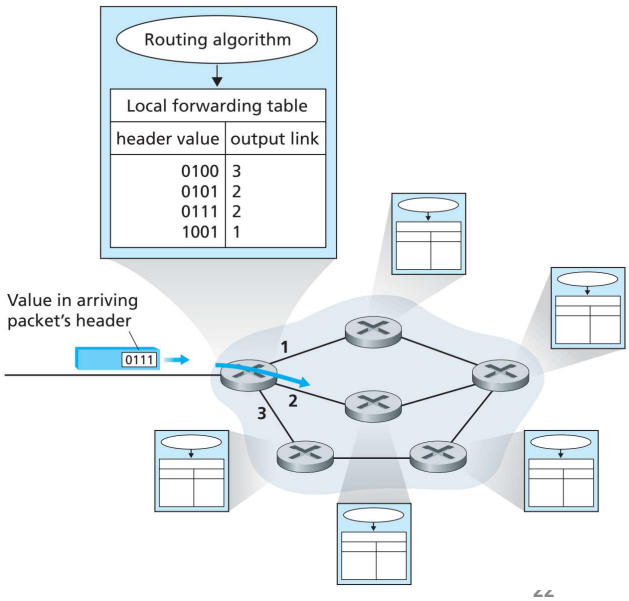
- **Forwarding**
  - **Move packets from a router's input link to appropriate router output link**
- **Routing**
  - **Determine route taken by packets from source to destination**
    - **Network layer must determine the path taken by packets while moving from sender to receiver**
      - **Routing Algorithms**

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Network Layer Function: Forwarding

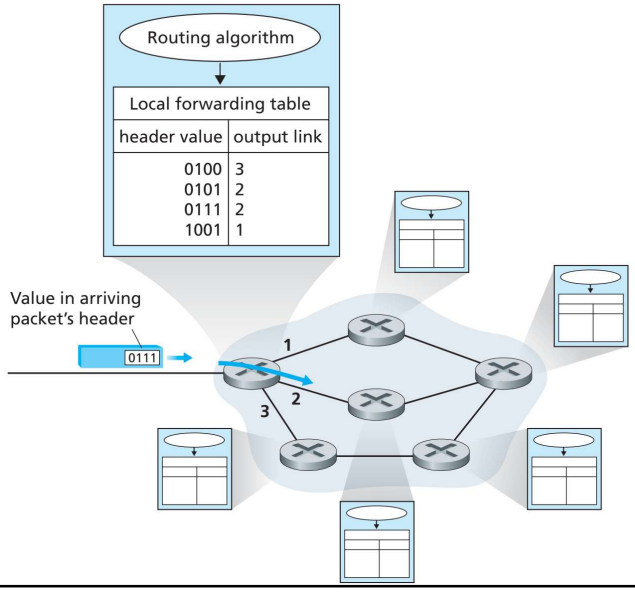
- **Move packets from a router's input link to appropriate router output link**
  - **Router's local action**
  - **Forwarding table at router**
    - Header field of incoming packet is used to index



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Network Layer Function: Forwarding



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Forwarding Table -Routers

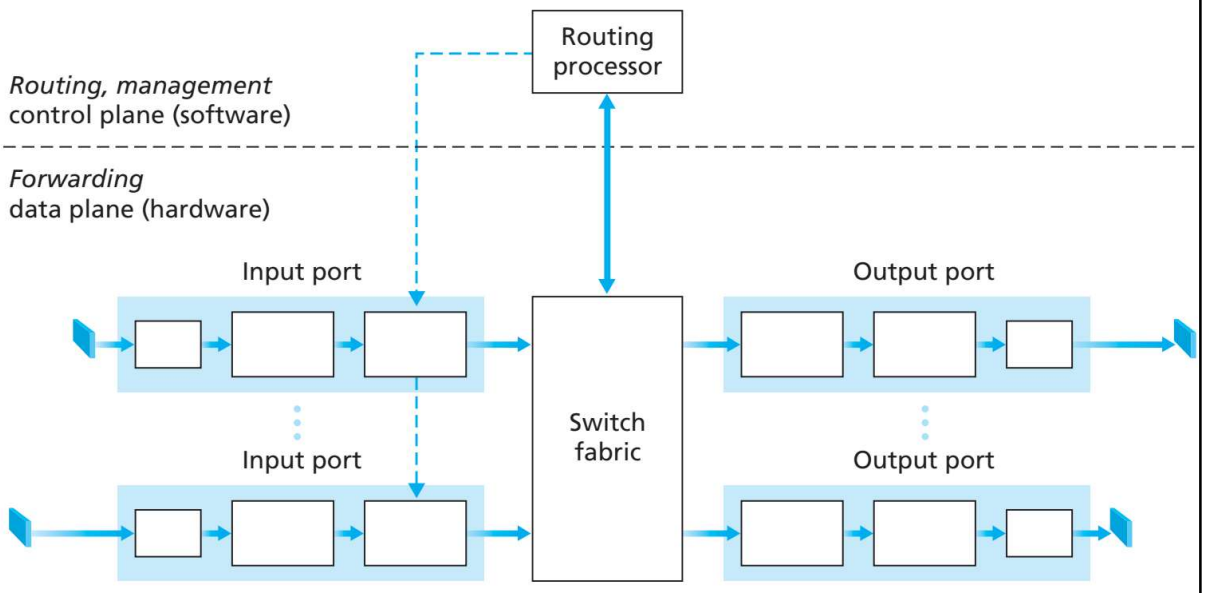
- **Configuring?**
  - **Routing algorithms decide what values to be written to these tables**
    - **Centralised– Runs at a central site**
    - **Decentralised – Runs at every router**

- **Routing**
  - **Determine route taken by packets from source to destination**
    - **Network layer must determine the path taken by packets while moving from sender to receiver**
      - **Routing Algorithms**

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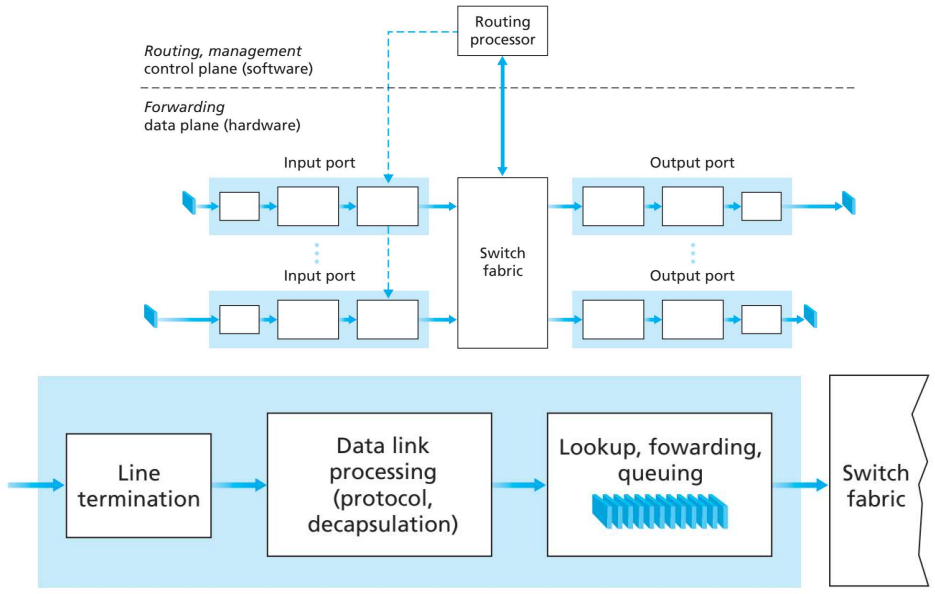
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Router Architecture



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Router Architecture-Input Port



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Decentralised Switching

- **Using header field values, lookup output port using forwarding table in input port memory ("match plus action")**
  - **Destination-based forwarding: forward based only on destination IP address (traditional)**
    - **One entry per destination in forwarding table**
    - **How many destination addresses possible?**
      - **Suppose 32 bit address-destination—4 billion possible addresses**
      - **Cannot store all the addresses**

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Destination based Forwarding

forwarding table

Destination Address Range	Link Interface
11001000 00010111 00010000 00000000 through 11001000 00010111 00010111 11111111	0
11001000 00010111 00011000 00000000 through 11001000 00010111 00011000 11111111	1
11001000 00010111 00011001 00000000 through 11001000 00010111 00011111 11111111	2
otherwise	3

- **No need to store 4 billion entries**

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Destination based Forwarding

- **Longest prefix matching**
  - **When looking for forwarding table entry for given destination address, use longest address prefix that matches destination address**

Destination Address Range	Link interface
11001000 00010111 00010*** ********	0
11001000 00010111 00011000 ********	1
11001000 00010111 00011*** ********	2
otherwise	3

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Destination based Forwarding

- **Longest prefix matching**

Destination Address Range	Link interface
11001000 00010111 00010*** ********	0
11001000 00010111 00011000 ********	1
11001000 00010111 00011*** ********	2
otherwise	3

examples:

11001000 00010111 00010110 10100001 which interface?

11001000 00010111 00011000 10101010 which interface?

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Destination based Forwarding

- **Longest prefix matching**

Destination Address Range	Link interface
11001000 00010111 00010*** ********	0
11001000 00010111 00011000 ********	1
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examples: 11001000 00010111 00010110 10100001 which interface?

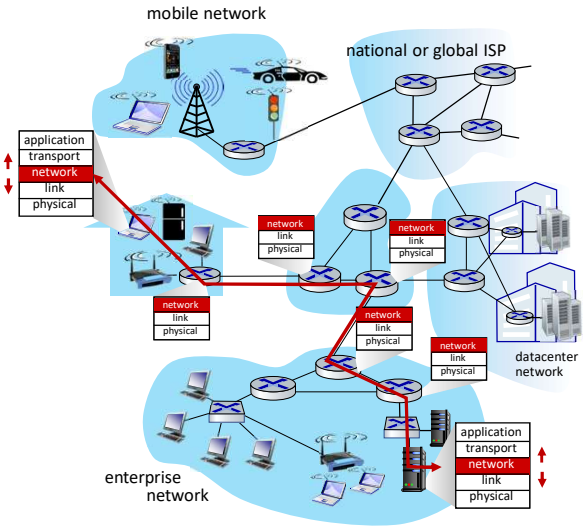
11001000 00010111 00011000 10101010 which interface?

- **Matching complexity?**
  - **Often performed using ternary content addressable memories (TCAMs)**

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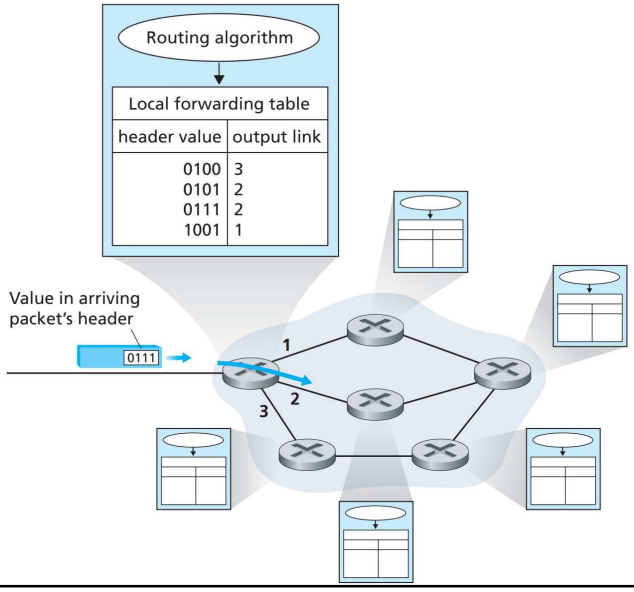
Network Layer



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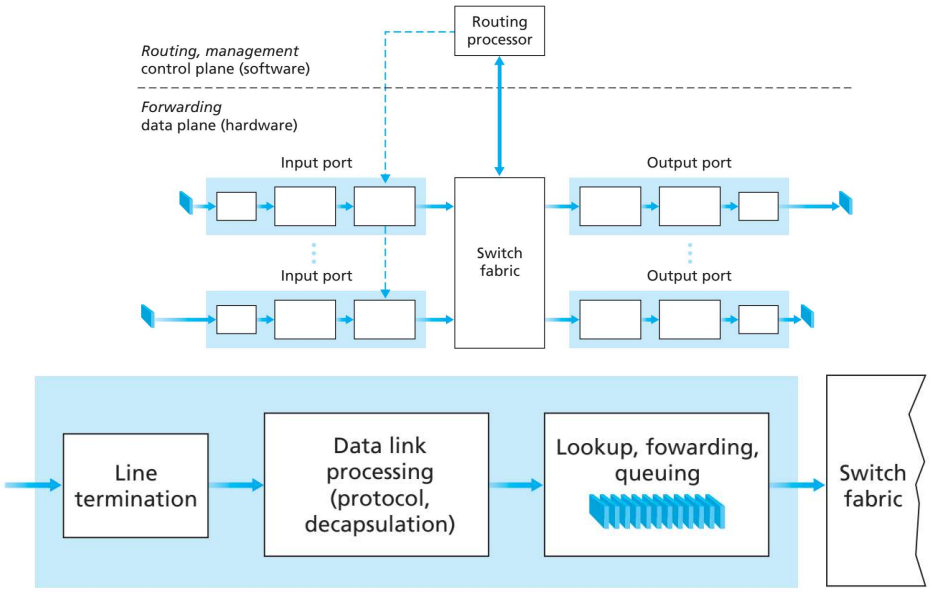
Network Layer Function: Forwarding



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Router Architecture-Input Port



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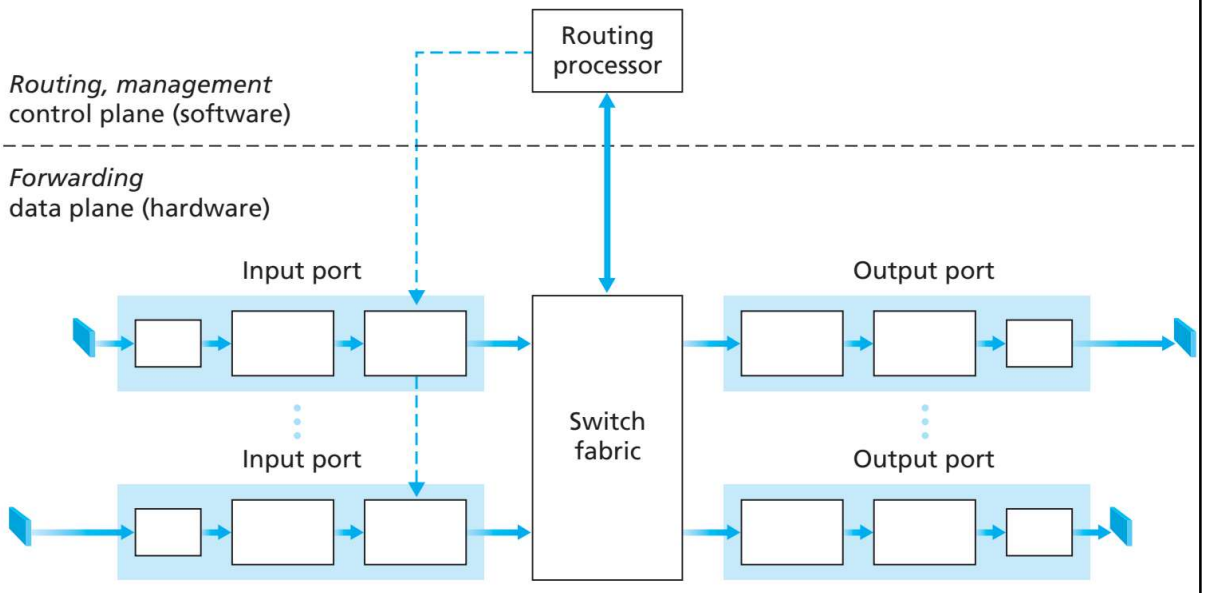
Input Port

- **Physical layer function of terminating an incoming link at a router**
- **Performs link-layer functions –interoperate with the link layer at the other side of the incoming link**
- **Lookup– Forwarding table**
  - **Destination based forwarding**
    - **Longest prefix matching**
  - **Most important responsibility**
  - **Passes the packet to switching fabric – mentioning the output link**
  - **May have to queued if there are other packets coming from**

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Router Architecture



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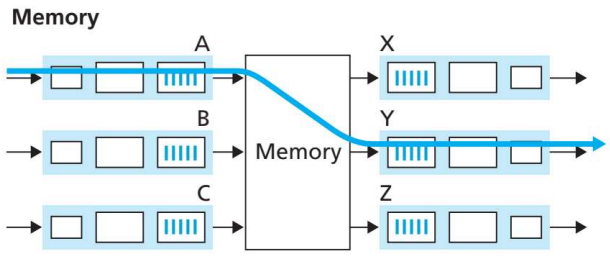
Switching Fabric

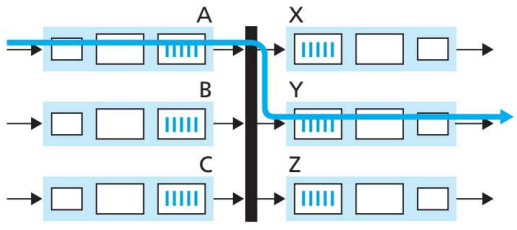
- Transfer packet from input link to appropriate output link
- Packets are actually switched
  - Forwarded
- Switching can be accomplished in different ways
  - Three major ways

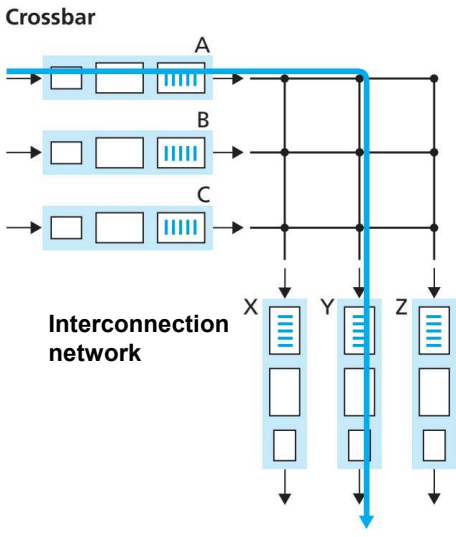
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Types of Switching





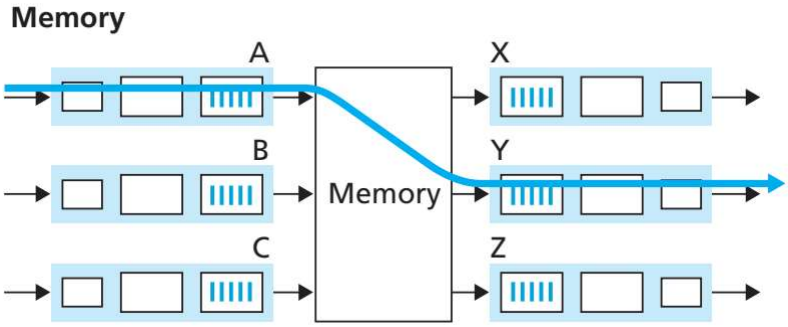


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Switching Via Memory

- **Earliest routers were traditional computers**
  - First generation routers
  - Switching between input and output ports under the control of CPU
- **Packet copied to system's memory**
- **Speed limited by memory bandwidth**

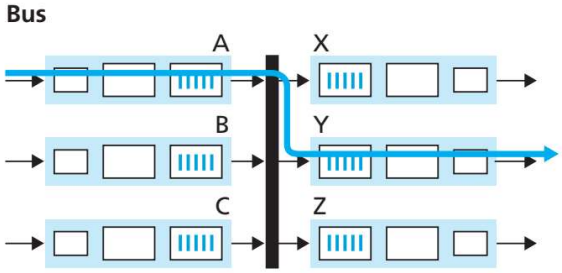


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Switching Via A Bus

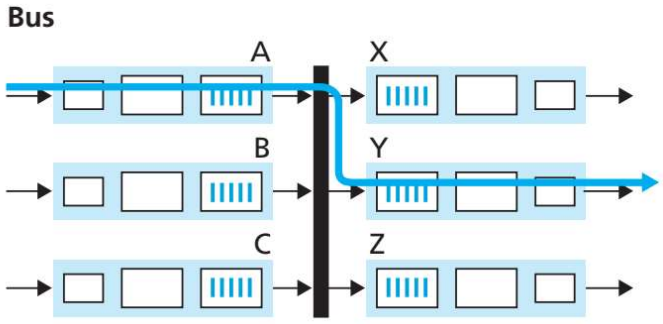
- **Datagram from input port memory to output port memory via a shared bus**
- **Input port to attach label-to indicate the output port**
- **The output port to which a packet is addressed receives the same**
  - Removes the label attached
- **Bus contention: switching speed limited by bus bandwidth**



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Switching Via A Bus

- **Bus contention:** switching speed limited by bus bandwidth
- Often sufficient for routers that operate in small local area and enterprise networks

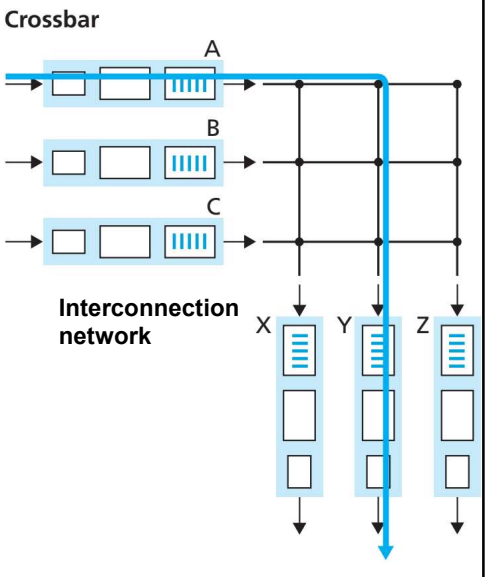


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Switching Via Interconnection Network

- To address the bandwidth issue faced due to single bus
- Use sophisticated interconnection networks
  - Used in the past to connect processors in multiprocessor setup
  - Crossbar switch - interconnection network
- 2N buses that connect N input ports to N output ports
  - 3x3 cross bar
- Vertical bus intersects horizontal bus at a crosspoint
  - Crosspoint can be opened or closed by switch fabric controller
- Multiple buses
  - Multiple packets can be switched in parallel

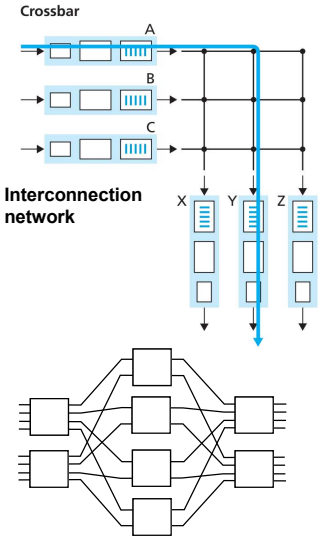


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Switching Via Interconnection Network

- **If 2 packets from 2 different input ports are destined to same output port**
  - **One must wait at the input port**
- **Scale and improving reliability**
  - **Multistage switch**
    - nxn switch from multiple stages of smaller switches
    - Connects inputs to outputs sequentially through multiple successive stages of smaller switching elements (like 2x2 crossbars)
  - **Multiple switching planes**



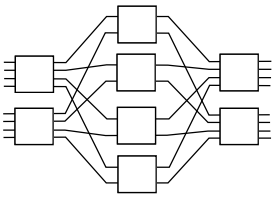
8x8 multistage switch
built from smaller-sized switches

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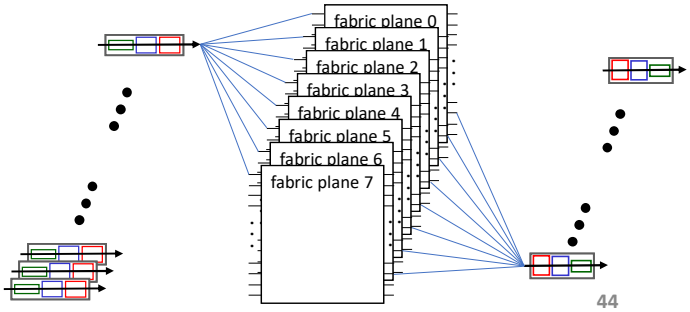
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Switching Via Interconnection Network

- **Scale and improving reliability**
  - Multistage switch
  - **Multiple switching planes**
    - **Uses parallel, independent copies of an entire switching fabric**
    - **Fragment datagram into fixed length cells on entry**
    - **Switch cells through the fabric, reassemble datagram at exit**
- **Cisco CRS router:**
  - Basic unit: 8 switching planes
  - Each plane: 3-stage interconnection network
  - Up to 100's Tbps switching capacity



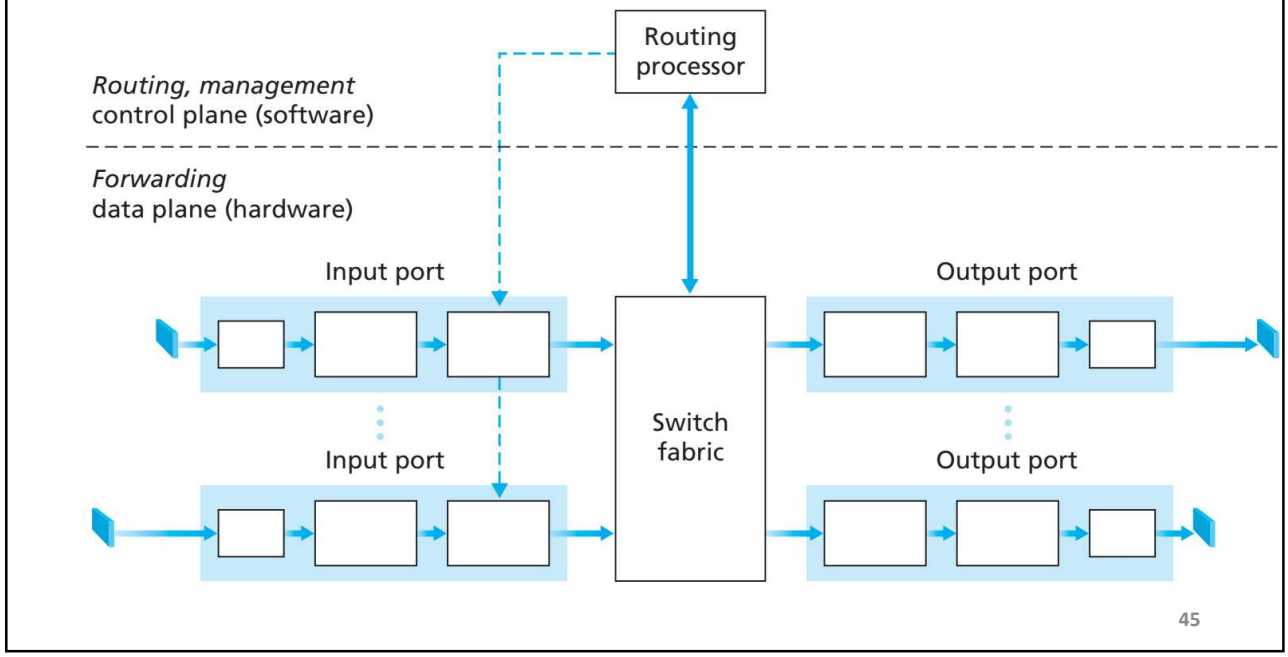
8x8 multistage switch
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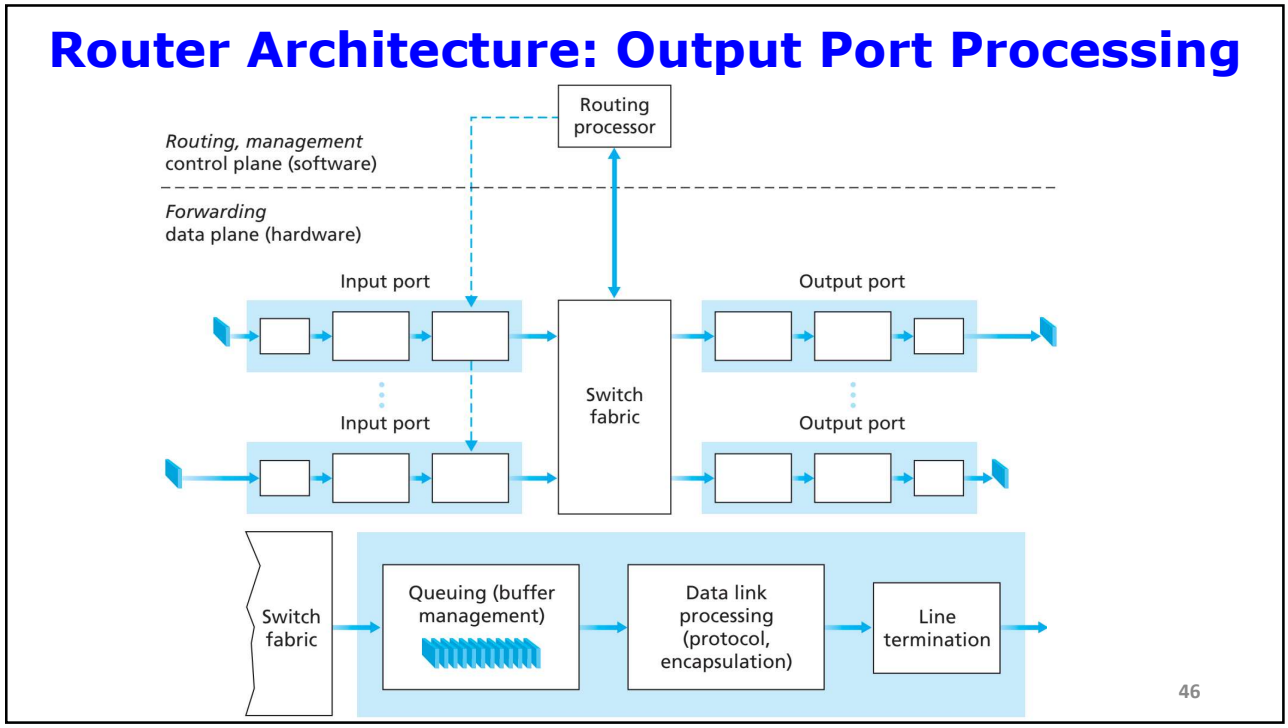
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Router Architecture



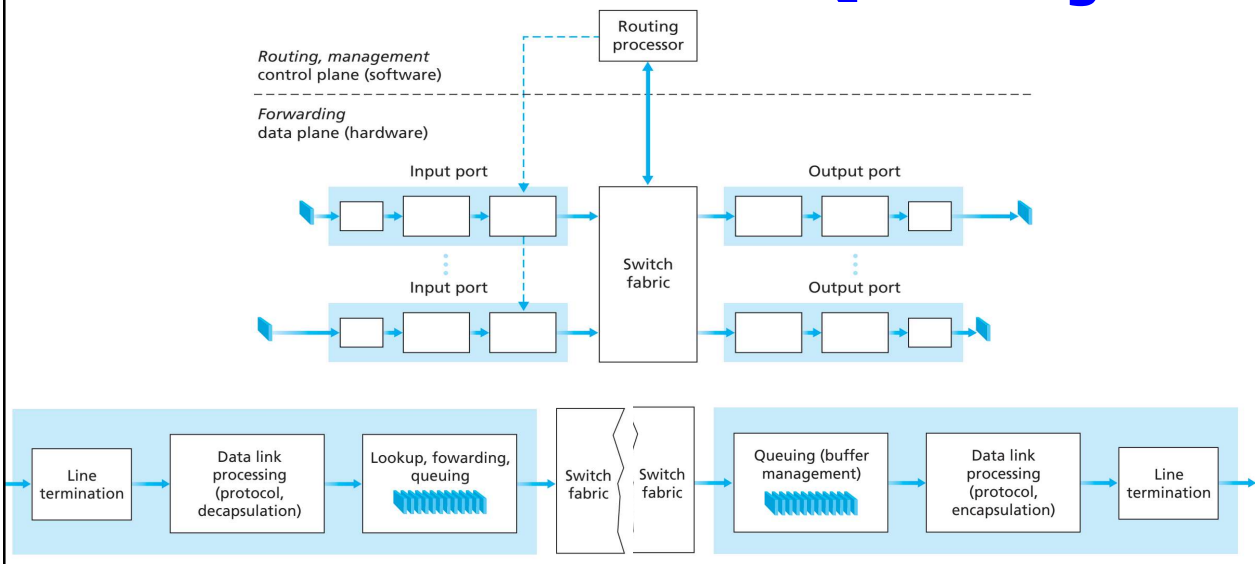
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Router Architecture: Output Port Processing



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Router Architecture: Queueing



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Router Architecture: Queueing

- **Both at input and output port**

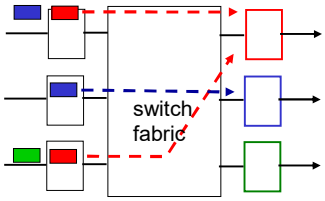
- **May result in packet loss**
  - **When the queue is full**
    - **Packets lost within the network**
    - **Packets dropped at router**

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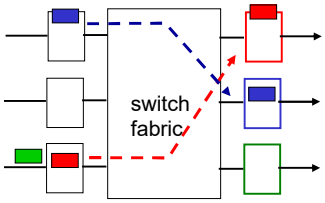
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Router Architecture: Input Port Queueing

- **If switch fabric slower than input ports combined**
- **Output port contention**
  - **When more than one datagram is destined to a single output port**
- **Head-of-the-Line (HOL) blocking**
  - **Queued datagram at front of queue prevents others in queue from moving forward**



output port contention: only one red datagram can be transferred. lower red packet is ***blocked***



one packet time later: green packet experiences HOL blocking

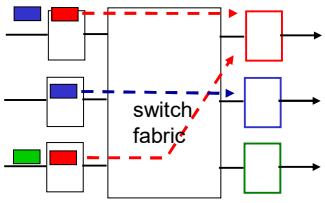
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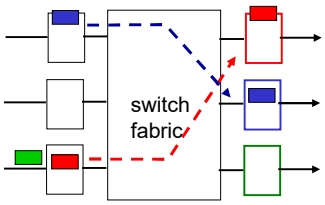
Router Architecture: Input Port Queueing

- **If switch fabric slower than input ports combined**
- **Output port contention**
  - **When more than one datagram is destined to a single output**
- **Head-of-the-Line blocking**
  - **Queued datagram at front of queue prevents others in queue from moving forward**

What is the kind of switching?



output port contention: only one red datagram can be transferred. lower red packet is ***blocked***



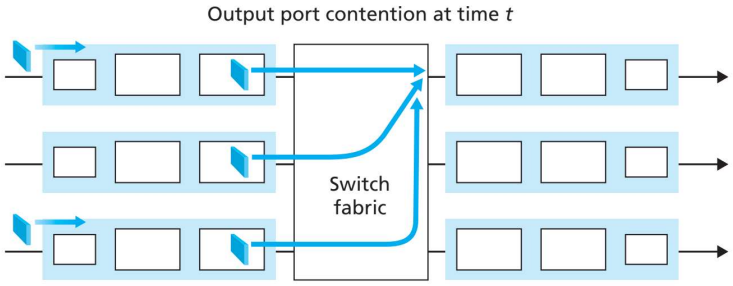
one packet time later: green packet experiences HOL blocking

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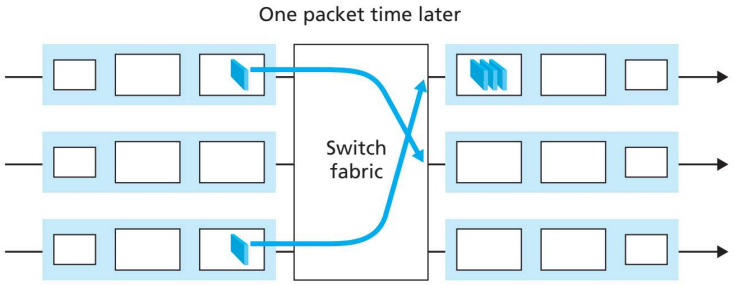
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Router Architecture: Output Port Queueing

- **Required when datagrams arrive from fabric faster than link transmission rate**



- **May result in packet/datagram loss**



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Router Architecture: Output Port Queueing

- **May result in packet/datagram loss**
  - **When the queue is full**
  - **Buffer size?**
- **How much buffering?**
  - **RFC 3439 rule of thumb:**
    - **average buffering, *B=RTT. C***
      RTT is average round trip time
      C is link capacity
    - **e.g., RTT of 250 msec and C = 10 Gbps link, B= 2.5 Gbit**
  - **Recent recommendations with N flows**

$$B = \frac{RTT.C}{\sqrt{N}}$$

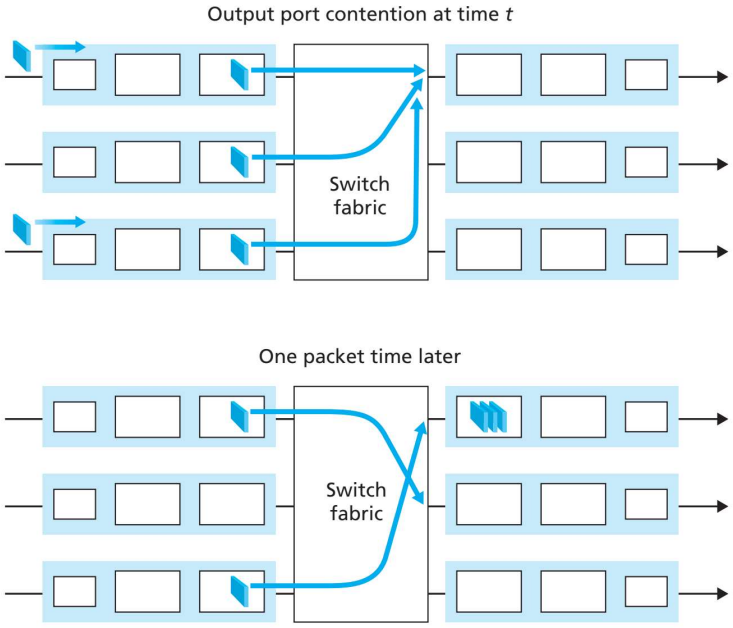
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Router Architecture: Output Port Queueing

- Required when datagrams arrive from fabric faster than link transmission rate
- May result in packet/datagram loss

- Packet scheduling
  - Choose among queued datagrams for transmission



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Output Port Queueing: Packet Scheduling

- Choose one packet from the queue to send over the link

- Different approaches
  - First come, first served (FCFS)
  - Priority
  - Round robin
  - Weighted fair queueing

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Packet Scheduling : FCFS

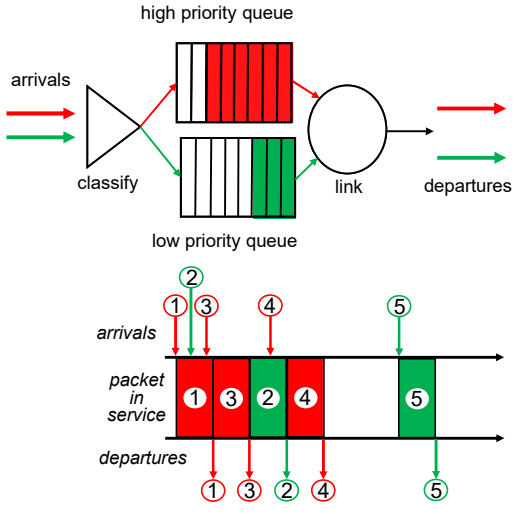
- First come, first served
- Simple approach
- Packets transmitted in order of arrival to output port
- Also known as: First-in-first-out (FIFO)
- Need more sophisticated scheduling
  - All end to end connections are given equal importnce

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Packet Scheduling : Priority

- Arriving traffic classified, queued by class
  - Any header fields can be used for classification
- Send packet from highest priority queue that has buffered packets
  - FCFS within priority class
- Issue?

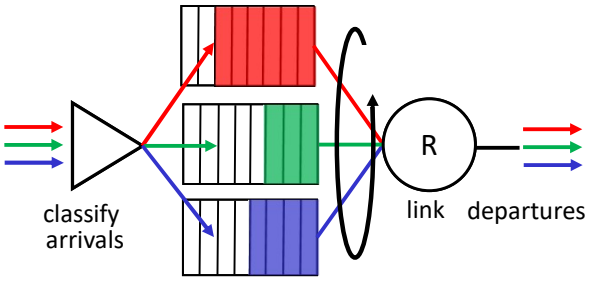


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Packet Scheduling : Round Robin

- Arriving traffic classified, queued by class
  - Any header fields can be used for classification
- Scheduler cyclically, repeatedly scans class queues, sending one complete packet from each class (if available) in turn
- Fair scheduling than Priority
- Priority class affected



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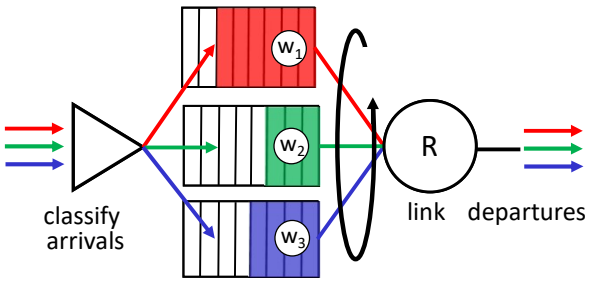
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Packet Scheduling : Weighted Fair Queueing

- Generalized round robin
- Arriving traffic classified, queued by class
  - Any header fields can be used for classification
- Each class, i, has weight, w_i, and gets weighted amount of service in each cycle:

$$\frac{w_i}{\sum_j w_j}$$

- Minimum bandwidth guarantee (per-traffic-class)
-



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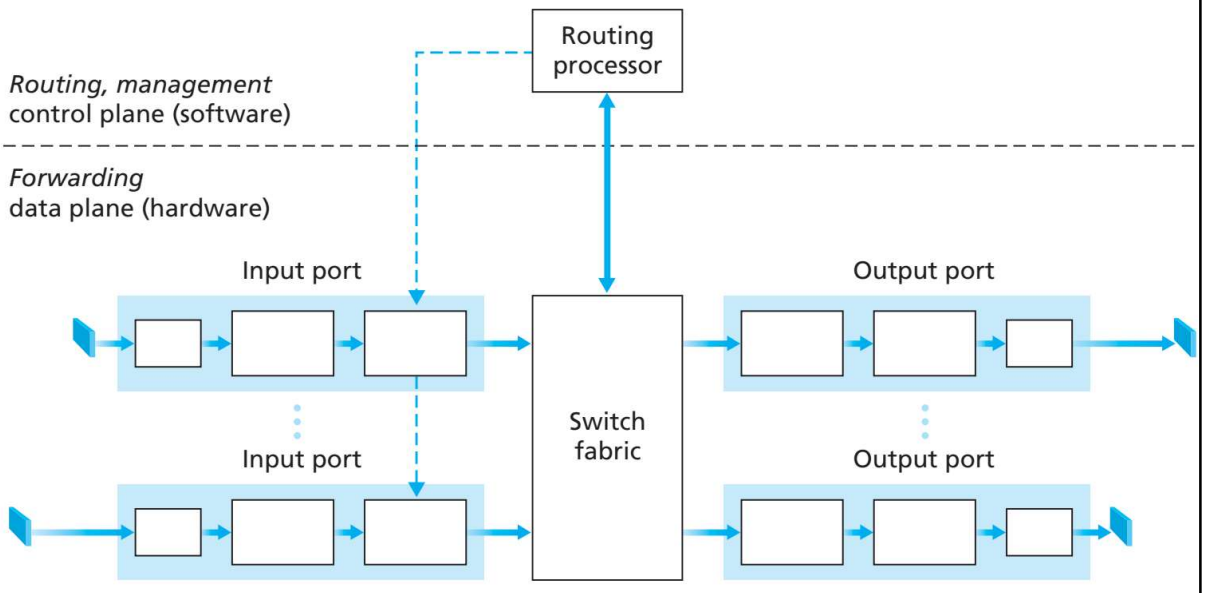
Input/Output Buffer Management

- **Buffer size – Not enough space for incoming packet**
  - **Suggested buffer sizes**
- **Still if there is no space left on buffers**
  - **May result in packet/datagram loss**
  - **Quality/reliability of service?**
- **Drop policy: which datagrams to drop if no free buffers?**
  - **Drop-tail: Drop the incoming packet when the buffer is full**
  - **Drop one or two existing packets to make room for the incoming packet**
    - **Priority based**
  - **Drop a packet before the buffer is full in order to provide a congestion signal to sender**
    - **Marking**
    - **Number of packet dropping and marking policies –Active queue management algorithms**
      - **Random early detection (RED)**
      - **Explicit congestion notification (ECN)**

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Router Architecture



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Router Architecture: Routing Processor

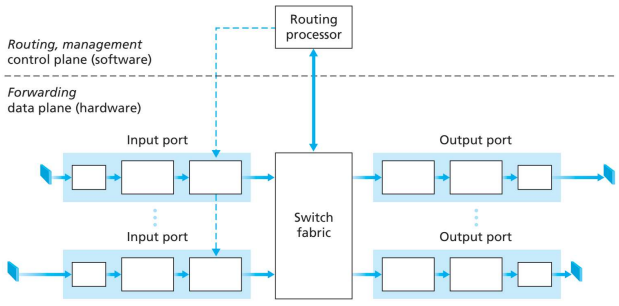
- Executes the routing protocols
- Maintains routing tables and attached link state information
- Computes the forwarding table for the router

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Router Architecture

- Routing control panel fully reside and execute in a routing processor
- Networkwide routing control panel is decentralized
  - Different pieces executing at different routers and communicating each others by sending messages
    - Routing algorithms
- Recent research – combination of centralized and decentralized



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